MARKING SCHEME

Model Paper — 01

Mathematics I

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Notes (for teachers):

- Part A: 2 marks per question (1 for a correct method, 1 for the correct final answer). Total 50.
- Part B: 10 marks per question. Total 50. The marks for each step are shown in the right-hand column.
- Give full marks for correct alternative methods. Accept reasonable ranges for values read from graphs and drawings.

Part A — Answers

1. Total work = 12 × 15 = 180 worker-days. Workers needed for 9 days = 180 ÷ 9 = 20. Extra workers = 20 − 12.

Final answer: 8 workers

(Marks 2)

2. Through E draw a line parallel to AB (and CD). Then AÊC = BÂE + ECD (alternate angles), so BÂE = 105° − 42°.

Final answer: BÂE = 63°

(Marks 2)

3. $\frac{3\times3 + 5\times2 - 1\times4}{12x} = \frac{9 + 10 - 4}{12x} = \frac{15}{12x}$

Final answer: $\frac{5}{4x}$

(Marks 2)

4. The side faces of a right prism are rectangles, so a rhombus that is not a square cannot be a face. (A square is possible when the lateral edge equals a base edge.)

Final answer: (ii) ticked

(Marks 2)

5. Students per degree = 12 ÷ 40 = 0.3. Cricket: 150° × 0.3 = 45. (or 12 × 150/40)

Final answer: 45 students

(Marks 2)

6. DC is the axis through the centre and the midpoint of AB, so DC ⊥ AB and DA = DB. ADBE is cyclic, so AD̂B = 180° − 110° = 70°. DC bisects AD̂B, so CD̂B = 70° ÷ 2.

Final answer: 35°

(Marks 2)

7. $3(x^2 - 25) = 3(x - 5)(x + 5)$

Final answer: 3(x − 5)(x + 5)

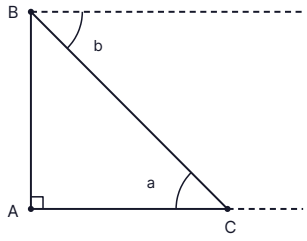
(Marks 2)

8. lg 50 = lg (5 × 10) = lg 5 + lg 10 = 0.6990 + 1

Final answer: 1.6990

(Marks 2)

9. Angle of elevation at C is a = ACB (between CA and CB). Angle of depression at B is b = angle between the horizontal through B and BC. They are alternate angles (the horizontals are parallel). See figure.



Final answer: $a = b$

(Marks 2)

10. Curved surface : base = $2\pi r(4r) : \pi r^2 = 8\pi r^2 : \pi r^2$

Final answer: 8 : 1

(Marks 2)

11. Total outcomes = 4 × 4 = 16. Sum 12: (4, 8), (6, 6), (8, 4) → 3 outcomes.

Final answer: $\frac{3}{16}$

(Marks 2)

12. DB = DC, so DB̂C = DĈB = (180° − 40°) ÷ 2 = 70°. AB̂D = 180° − 70° = 110°. AB = BD, so x = (180° − 110°) ÷ 2.

Final answer: x = 35°

(Marks 2)

13. A = {1, 2, 3, 4, 6}, B = {2, 4, 6, 8, 10}. A ∩ B = {2, 4, 6} so (i) is false; A ∪ B = {1, 2, 3, 4, 6, 8, 10} so (ii) is true; A′ = {5, 7, 8, 9, 10}, n(A′) = 5 so (iii) is false.

Final answer: (ii) ticked

(Marks 2)

14. $\frac{30x^4y^5}{10x^2y^2}$

Final answer: $3x^2y^3$

(Marks 2)

15. Common ratio $r = 6 \div (-12) = -\frac{1}{2}$. Fifth term $= 6 \times r^2 = 6 \times \frac{1}{4}$

Final answer: $\frac{3}{2}$ (1.5)

(Marks 2)

16. $B\hat{A}C = B\hat{D}C = 29°$ (same segment). So $A\hat{D}C = 2 \times 29° = 58°$. $C\hat{B}E = A\hat{D}C$ (exterior angle of a cyclic quadrilateral).

Final answer: 58°

(Marks 2)

17. HCF of 24, 36, 60 = 12; lowest power of p is p^2; lowest power of q is q.

Final answer: $12p^2q$

(Marks 2)

18. Volume of prism $= 154 \times h = \pi r^2 h \Rightarrow r^2 = 154 \times \frac{7}{22} = 49$

Final answer: $r = 7$ cm

(Marks 2)

19. D and E are midpoints, so BD = AD = 3 and EC = AE = 5. In the right-angled triangle ADE, $DE = \sqrt{(5^2 - 3^2)} = 4$. By the midpoint theorem $BC = 2 \times DE = 8$. Perimeter = BD + DE + EC + CB = 3 + 4 + 5 + 8.

Final answer: 20 cm

(Marks 2)

20. Gradient $= (5 - 3) \div (6 - 2) = \frac{1}{2}$. $y - 3 = \frac{1}{2}(x - 2)$

Final answer: $y = \frac{1}{2}x + 2$ (or $x - 2y + 4 = 0$)

(Marks 2)

21. In a parallelogram opposite angles are equal: $D\hat{A}B = B\hat{C}D = 70°$. ADEB is a cyclic quadrilateral, so $D\hat{E}B = 180° - 70°$.

Final answer: 110°

(Marks 2)

22. $P \times (1.05)^2 = 22\,050$, $P \times 1.1025 = 22\,050$, $P = 22\,050 \div 1.1025$

Final answer: Rs 20 000

(Marks 2)

23. CB = CD (radii) so $C\hat{B}D = C\hat{D}B = 70°$. AC ∥ BD, so $A\hat{C}B = C\hat{B}D = 70°$ (alternate angles). The angle at the centre $A\hat{C}B = 70°$, so $A\hat{E}B = 70° \div 2$.

Final answer: 35°

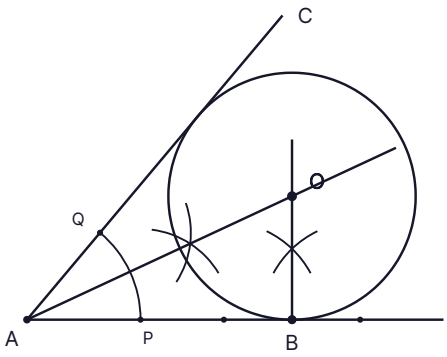
(Marks 2)

24. $4x + 15 > 3 \Rightarrow 4x > -12 \Rightarrow x > -3$

Final answer: x > −3; negative integers: −2, −1

(Marks 2)

25. Bisector of $B\hat{A}C$ (complete the arcs from P and Q, join A to their intersection and produce); perpendicular to AB at B (arcs on either side of B on AB, then two arcs above); O is where they meet. See figure.



The completed construction (shown reduced in size). Accept the circle drawn or not.

Final answer: O = intersection (OB = distance from O to AC)

(Marks 2)

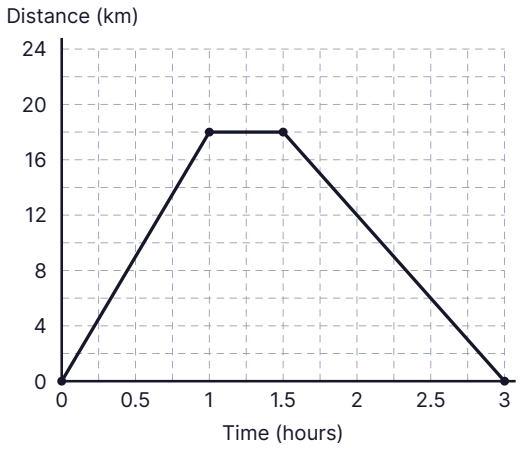
Part B — Answers

Question 1

(i)	
Fraction left after food = $1 - \frac{1}{5} = \frac{4}{5}$	**1**
(ii)	
Other expenses = $\frac{5}{8} \times \frac{4}{5} = \frac{1}{2}$ of the money received	**1**
Left = $\frac{4}{5} - \frac{1}{2} = \frac{3}{10}$	**1**
(iii)	
$\frac{3}{10}$ of the money received = Rs 36 000, so money received = $36\,000 \times \frac{10}{3}$ = Rs 120 000	**1**
Food = $\frac{1}{5} \times 120\,000$ = Rs 24 000; other expenses = $\frac{1}{2} \times 120\,000$ = Rs 60 000	**1**
(iv)	
Let the salary be Rs S. Tax = $\frac{10}{100} \times (S - 75\,000)$. $S - 0.1(S - 75\,000) = 120\,000$	**1**
$0.9S + 7\,500 = 120\,000$, so $0.9S = 112\,500$	**1**
S = Rs 125 000 (check: tax = 10% of 50 000 = 5 000; 125 000 − 5 000 = 120 000)	**1**
(v)	
$\frac{24000}{125000} \times 100$	**1**
= 19.2%	**1**

Total marks: 10

Question 2

(a)(i)	
 A correct graph (shown reduced in size)	
Horizontal line from (1, 18) to (1.5, 18) (stay of 30 minutes)	**1**
Point (3, 0) correctly located (return takes 1½ h, so reaches home at 3 h)	**1**
Straight line from (1.5, 18) to (3, 0)	**1**
(a)(ii)	
Speed = distance ÷ time = 18 ÷ 1.5	**1**
= 12 km h^{-1}	**1**
(b)(i)	
Arc AEB = $\frac{1}{2} \times 2 \times \frac{22}{7} \times 7 = 22$ cm	**1**
Perimeter = 22 + 33 + 14 + 33 = 102 cm	**1**
(b)(ii)	
Area of semicircle = $\frac{1}{2} \times \frac{22}{7} \times 7 \times 7 = 77$ cm^2	**1**
Area of rectangle = 14 × 33 = 462 cm^2	**1**
462 ÷ 77 = 6 times	**1**

Total marks: 10

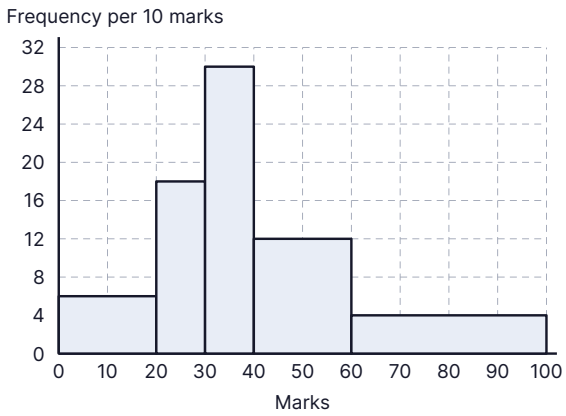
Question 3

(i)	
Rates = $\frac{10}{100} \times 50\,000$	**1**
= Rs 5 000	**1**
(ii)	
Rent for the year = 12 500 × 12 = Rs 150 000; maintenance = 10% of 150 000 = Rs 15 000	**1**
Money left = 150 000 − 5 000 − 15 000 = Rs 130 000	**1**
(iii)(a)	
Dividend per share = $\frac{7.5}{100} \times 80$ = Rs 6	**1**
Number of shares = 15 000 ÷ 6	**1**
= 2 500 shares	**1**
(iii)(b)	
Total invested = 2 500 × 80 = Rs 200 000	**1**
Extra money added = 200 000 − 130 000 = Rs 70 000	**1**
Percentage = $\frac{70000}{200000} \times 100 = 35\%$	**1**

Total marks: 10

Question 4

(i)	
Values read from the polygon: 20 – 30 → 18; 30 – 40 → 30; 40 – 60 → 24; 60 – 100 → 16 (all four correct 3; three correct 2; two correct 1)	**3**
(ii)	
Heights (frequency per 10 marks): 20 – 30: 18 ÷ 10 × 10 = 18; 30 – 40: 30; 40 – 60: 24 ÷ 20 × 10 = 12; 60 – 100: 16 ÷ 40 × 10 = 4 (first bar: 12 ÷ 20 × 10 = 6, given)	**1**

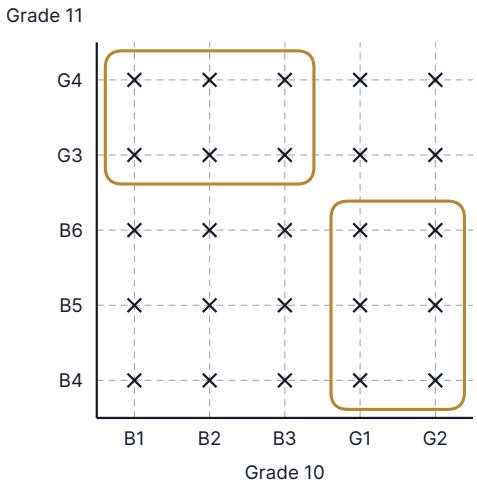


A correct histogram (shown reduced in size)

Bars for 20 – 30 and 30 – 40 drawn with heights 18 and 30	**1**
Bars for 40 – 60 and 60 – 100 drawn with heights 12 and 4, bar widths correct and no gaps between bars	**1**
(iii)	
Cumulative frequencies: 12, 30, 60, 84, 100 (12 + 18 = 30; 30 + 30 = 60; 60 + 24 = 84; 84 + 16 = 100) (all four correct 2; two or three correct 1)	**2**
(iv)	
Students with 40 or more = 24 + 16 = 40 (or 100 − 60)	**1**
Probability = $\frac{40}{100} = \frac{2}{5}$	**1**

Total marks: 10

Question 5

(i)	
 Sample space and the enclosed outcomes (shown reduced in size)	
All 25 points (5 columns × 5 rows) marked with '×' at the grid intersections (24 or 25 correct: 2; at least 13 correct: 1)	**2**
(ii)	
Closed curves around the 6 outcomes (B1, B2, B3) × (G3, G4) and the 6 outcomes (G1, G2) × (B4, B5, B6); 12 outcomes in all	**2**
P(exactly one boy) = $\frac{12}{25}$	**1**
(iii)	
First branches: $\frac{12}{25}$ (exactly one boy) and $\frac{13}{25}$ (both boys: 9, both girls: 4, so 13 outcomes)	**1**
After 'exactly one boy': wins $\frac{3}{5}$, does not win $\frac{2}{5}$	**1**
After 'both boys or both girls': wins $\frac{2}{5}$, does not win $\frac{3}{5}$	**1**
(iv)	
P(does not win) = $\frac{12}{25} \times \frac{2}{5} + \frac{13}{25} \times \frac{3}{5} = \frac{24}{125} + \frac{39}{125}$	**1**
= $\frac{63}{125}$	**1**

Total marks: 10